LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Unit Test: Sound

Chapter / Topic: Sound (topic-based)

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SECTION A

Q1. Sound travels fastest in: **[1]**

Answer: (c) steel

Q2. The SI unit of frequency is the: **[1]**

Answer: (d) hertz

Q3. Sound waves in air are: **[1]**

Answer: (b) longitudinal

Q4. Sound of frequency higher than 20 kHz is called: **[1]**

Answer: (b) ultrasonic

Q5. A sound has a frequency of 500 Hz. Its time period is: **[1]**

Answer: (a) 0.002 s (T = 1/f)

Q6. **Assertion (A):** We see a lightning flash before we hear the thunder. **Reason (R):** Light travels much faster than sound. **[1]**

Answer: (a)

SECTION B

Q7. What are compressions and rarefactions? **[2]**

Answer: When sound travels through air, compressions are regions where the particles are crowded together (high pressure), and rarefactions are regions where they are spread apart (low pressure).

- Compression: 1
- Rarefaction: 1

OR, answer: Reverberation is the persistence of sound in a big hall due to repeated reflections from the walls, ceiling and floor. It can be reduced by covering these surfaces with sound-absorbing materials such as fibreboard, rough plaster or curtains, and by using seats made of absorbent material.

- Definition: 1
- How to reduce: 1

Q8. Sound travels at 340 m/s in air. Find the frequency of a sound wave of wavelength 1.7 m. **[2]**

Answer: $f = v/\lambda = 340/1.7 = 200$ Hz

- Formula: 1
- 200 Hz: 1

SECTION C

Q9. A boy claps his hands in front of a tall building and hears the echo 3 s later. (Speed of sound = 340 m/s) **[3]**

(i) $340 \times 3 = 1020$ m [1]

(ii) $1020/2 = 510$ m (the sound goes there and back) [1]

(iii) The echo must return after at least 0.1 s, so the distance = $(340 \times 0.1)/2 = 17$ m [1]

Q10. Give three uses of ultrasound. **[3]**

Answer: (Any three) Cleaning hard-to-reach parts of objects such as spiral tubes and electronic components; detecting cracks and flaws in metal blocks; imaging the heart (echocardiography) and

other internal organs (ultrasonography); breaking kidney stones into fine grains; SONAR, for finding the depth of the sea and locating underwater objects.

- Any three: 1 each

OR, answer: SONAR (Sound Navigation And Ranging) uses ultrasonic waves to measure the distance, direction and speed of underwater objects. A transmitter on a ship sends ultrasonic waves down. They reflect from the sea bed and are picked up by a detector. If the echo takes time t and the speed of sound in water is v, the depth = (v × t)/2.

- What SONAR is: 1
- Working: 1
- Depth = vt/2: 1

SECTION D

Q11. Answer the following: **[5]**

(i) When an object vibrates forward, it pushes the air particles in front of it and creates a region of high pressure (compression). When it moves back, it creates a region of low pressure (rarefaction). As it keeps vibrating, a series of compressions and rarefactions travels through the air. The air particles themselves only vibrate about their positions; they do not travel with the wave. The disturbance reaches our ear and makes the eardrum vibrate. [3]

- Compression formed: 1
- Rarefaction formed: 1
- Disturbance travels, particles don't: 1

(ii) Depth = (v × t)/2 = (1500 × 4)/2 = 3000 m [2]

- Formula with ÷2: 1
- 3000 m: 1

SECTION E

Q12. The outer ear collects sound and sends it through the ear canal to the eardrum, which starts vibrating. In the middle ear, three tiny bones (the… **[4]**

(i) The eardrum. [1]

(ii) Hammer, anvil and stirrup. [1]

(iii) The cochlea converts the vibrations (pressure variations) into electrical signals that are sent to the brain. Very loud sounds can damage the eardrum and the inner ear, causing temporary or permanent hearing loss. [2]

- Cochlea's role: 1
- Harm from loud sounds: 1